

# Questions: Introduction to data analysis

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## Summary

A selection of questions for the study guide on introduction to data analysis.

*Before attempting these questions, it is highly recommended that you read [Guide: Introduction to data analysis](#).*

## Q1

Cantor's Confectionery records how many customers had entered its store for the past 6 days. This can be summarized by the following list:  $\{17, 91, 23, 18, 27, 22\}$ .

- 1.1. Find the mean of this data set.
- 1.2. Find the median of this data set.
- 1.3. Why would it be important to find the median of this data set, instead of only finding the mean?
- 1.4. Why would it not be important to find the mode of this data set?
- 1.5. Can you think of any potential reasons for the existence of the outlier?

## Q2

For each data set below, find the mean, median, mode (if possible), range, and interquartile range.

- 2.1.  $\{14, 71, 82, 39, 39, 71, 71, 48\}$
- 2.2.  $\{919, 3293, 912, 7775\}$
- 2.3.  $\{3, 72, 800, 8, 763, 8, 3, 9028, 763, 39, 3\}$

## Q3

Examine the following scenarios, and identify which type of data visualization would be most suitable for each scenario. Your options include: bar chart, line graph, histogram, pie chart, and scatter plot.

- 3.1. A business wants to visualize how the budget allocations breakdown of each department compares to the totality of the budget.
- 3.2. A florist wants to visualize the distribution of flower stem lengths.
- 3.3. A teacher wants to visualize their students' exam scores against the number of hours they studied.
- 3.4. A bakery wants to visualize the trends in its chocolate bread sales revenue over time.

## Q4

For each scenario below, determine which of the following options would be most appropriate to use: correlation test, regression analysis, confidence interval, PDF, PMF, and CDF.

- 4.1. A researcher wants to test whether there is a relationship between cholesterol levels and heart disease risk.
- 4.2. A researcher wants to model the relationship between cholesterol levels and heart disease risk in terms of a linear function.
- 4.3. You flip a coin three times. Each flip has a 0.5 chance of the coin landing on 'tails'. You know that this can be modelled as a Binomial distribution, and you want to find the probability that the coin lands on 'tails' all three times.
- 4.4. A biologist collects a random sample of 100 birds and calculates their mean wingspan. They want to find a range of values in which the mean wingspan of all birds lies, with 90% confidence.

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[After attempting the questions above, please click this link to find the answers.](#)

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## Version history and licensing

v1.0: initial version created 9/25 by Michelle Arnetta as part of a University of St Andrews VIP project.

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